

Risk Management Analysis

Scenario 1: Hedging Market Risk

Situation:

Coffee prices are expected to rise due to adverse weather, posing cost risk for buyers (e.g., roasters).

Risk Management Options:

1. **Buy Coffee Futures Contracts**
 - Lock in current prices using the **cost of carry model** to determine fair value.
 - This strategy secures input costs and removes price uncertainty.
2. **Buy Call Options on Coffee**
 - Use the **Black-Scholes model** to price options.
 - Provides upside protection (locks in maximum purchase price) while retaining benefit if prices fall.
3. **Implement a Collar Strategy**
 - Buy a call option and simultaneously sell a put option to reduce net premium cost.
 - Effective in balancing protection with limited flexibility.

Scenario 2: Credit Risk with a New Supplier

Situation:

Entering a contract with a supplier in a volatile economic region—high risk of payment/default issues.

Risk Management Options:

1. **Credit Scoring and Evaluation**
 - Use models like **logistic regression** or **Altman Z-score** to assess default probability.
2. **Require Collateral or Prepayment**
 - Mitigate loss exposure by requiring margin or partial payment up front.
 - Use **Potential Future Exposure (PFE)** models to determine collateral amount.
3. **Purchase Credit Default Swap (CDS)**
 - Hedge against supplier default by transferring risk to a third party.
 - Price CDS using expected loss and credit spread models.

Scenario 3: Operational Risk in Trade Execution

Situation:

Manual trading processes are leading to frequent errors and increased operational losses.

Risk Management Options:

1. **Implement Automated Trading Algorithms**
 - Reduce human error using pre-programmed strategies (e.g., moving average crossovers).
 - Backtest and monitor using performance metrics.
2. **Enhance IT Infrastructure and Monitoring**
 - Upgrade systems for faster execution and fewer outages.
 - Model system reliability using **downtime probability models**.
3. **Conduct Regular Training and Performance Reviews**
 - Train staff on risk systems and trade protocols.
 - Measure effectiveness via error rate reduction and post-training assessments.